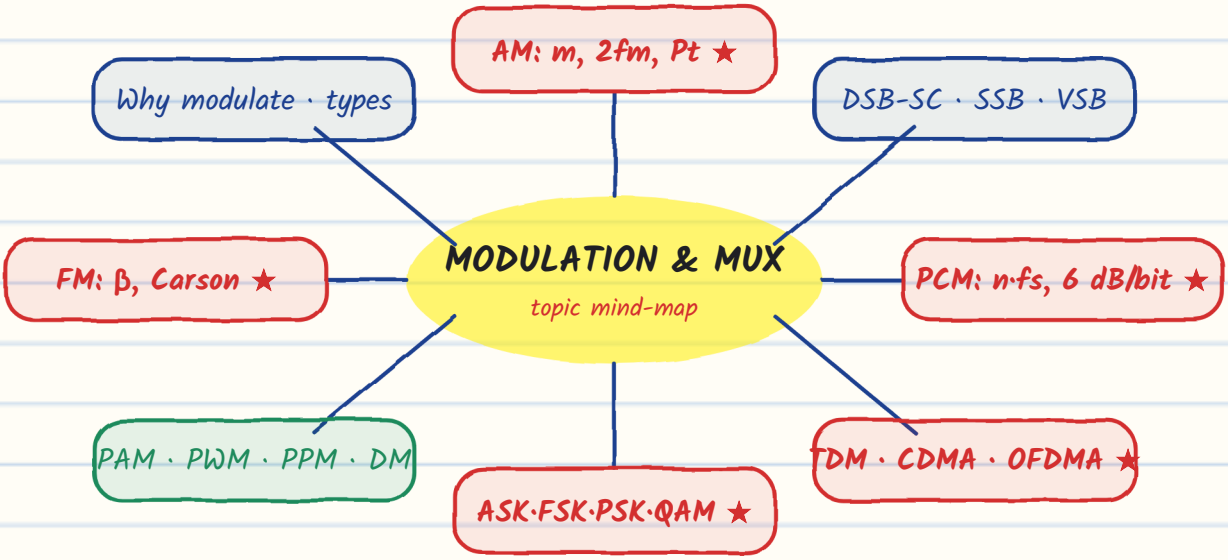
Modulation & Multiplexing

AM · FM · PCM · DM · keying · TDM · CDMA: exam-ready in 12 sheets



↑ the whole lesson on one page: red = high-yield, green = method / applied

➜ Syllabus coverage

"Analog communication system: different modulations, modulation circuits and detectors; Digital modulation: different standard modulation schemes; Pulse and digital signals: PAM, PCM, DM, ADM; Multiplexing: TDM, FDM, multiple-access network: TDMA, FDMA, CDMA".

High-yield list

① $P_t = P_c(1 + m^2/2)$
② BW: AM $2f_m$, SSB f_m
③ Carson $2(\Delta f + f_m)$
④ PCM $R_b = n \cdot f_s$
⑤ SQNR ≈ 6.02n + 1.76 dB
⑥ bits/symbol = $\log_2 M$
⑦ T1 1.544, E1 2.048 Mbps
⑧ GSM = FDMA + TDMA

Legend yellow = key term · red circle = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

➜ Contents

✓ Why modulate · AM waveform ..... p. 2
✓ AM power · DSB-SC/SSB/VSB ..... p. 3
✓ Superhet · circuits · FM/PM ..... p. 4
✓ AM vs FM · PAM/PWM/PPM ..... p. 5
✓ PCM p. 6 · DM / ADM / DPCM ... p. 7
✓ Line codes · ASK/FSK/PSK ..... p. 8
✓ M-ary, QAM, OFDM ..... p. 9
✓ Multiplexing & multiple access ... p. 10
✓ Night-before sheet ..... p. 11
✓ Practice MCQs + written ..... p. 12

1 Why Modulate? MCQ + Written

Definition **Modulation** = varying the amplitude, frequency or phase of a high-frequency carrier in step with the low-frequency message (baseband) signal. **Demodulation (detection)** recovers the message at the receiver.

1. **Antenna size:** an antenna is about λ/4 long, λ = c/f
2. **Multiplexing:** many stations on different carriers (FDM)
3. **Range:** RF radiates and propagates far better than audio
4. **Noise trade-off:** FM and spread spectrum buy SNR with bandwidth

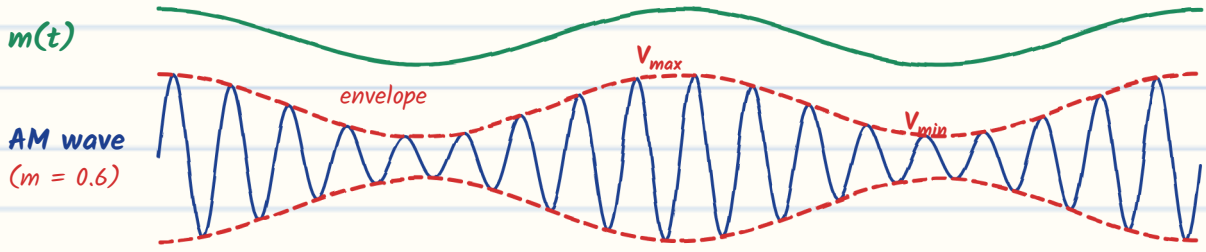
Worked example ✍ Antenna for a 1 kHz tone: λ = 3×10⁸ / 10³ = 300 km → λ/4 = 75 km (impossible). On a 1 MHz carrier: λ/4 = 75 m; at 100 MHz (FM band): **0.75 m**.

Memory trick The name says what varies: **A**mplitude, **F**requency, **P**hase; PAM/PWM/PPM = pulse **A**mplitude / **W**idth / **P**osition. FM + PM = angle modulation.

Exam trap PCM and DM are **A/D (source-coding) methods**, not carrier modulations. Their bits go out baseband (line code) or on a carrier (ASK/FSK/PSK).

2 Amplitude Modulation (AM) high-yield

Definition The carrier **amplitude** follows the message; frequency and phase stay fixed. Single tone: $s(t) = A_c[1 + m\cos 2\pi f_m t]\cos 2\pi f_c t$, modulation index $m = A_m/A_c$ ($m \times 100\%$ = percentage modulation).

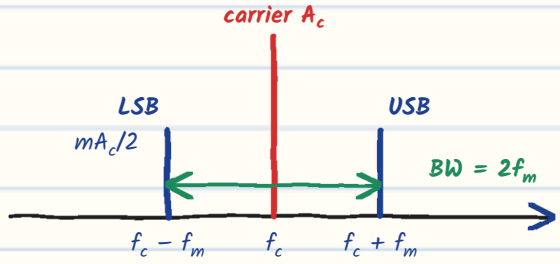


the envelope (red dashes) is a copy of the message: that is why a simple diode detector works

From a CRO trace: $m = (V_{max} - V_{min})/(V_{max} + V_{min})$. Several tones: $m_t = \sqrt{(m_1^2 + m_2^2 + \ldots)}$. General message: $s(t) = [A_c + k_a m(t)]\cos 2\pi f_c t$, envelope correct only while $1 + k_a m(t) \geq 0$.

Worked example ✍ V_{max} = 10 V, V_{min} = 2 V → m = 8/12 = **0.67 (67%)**. Tones $m_1 = 0.3$, $m_2 = 0.4$ → $m_t = \sqrt{(0.09 + 0.16)} = 0.5$.

→ AM spectrum: carrier + two sidebands



Expand with cos A cos B: $s(t) = A_c \cos \omega_c t + (mA_c/2)\cos(\omega_c + \omega_m)t + (mA_c/2)\cos(\omega_c - \omega_m)t$

→ carrier + **upper** + **lower sideband**; bandwidth **BW = $2f_m$** (highest message frequency).

f_c = 1 MHz, f_m = 5 kHz → LSB 995 kHz, USB 1005 kHz, BW = **10 kHz**.

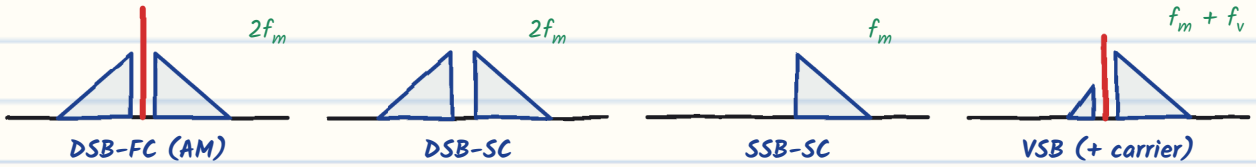
m	Name	Result
$m < 1$	under-modulation	normal; envelope = message
$m = 1$	100% modulation	largest undistorted depth; $V_{min} = 0$
$m > 1$	over-modulation	envelope distortion + splatter (extra sidebands)

Power (resistance R) $P_c = A_c^2/2R$ · each sideband = $\mathbf{m^2P_c/4}$ · $P_{SB} = m^2P_c/2$

$P_t = P_c(1 + m^2/2)$ $I_t = I_c\sqrt{(1 + m^2/2)}$ $\eta = P_{SB}/P_t = m^2/(2 + m^2) \le$ **33.3%** (at $m = 1$, the carrier still takes 2/3)

Worked examples ✍️ (a) P_c = 1 kW, $m = 0.6 \rightarrow P_t = 1000 \times 1.18 =$ **1180 W**, P_{SB} = 180 W (90 W each), $\eta = 180/1180 =$ **15.3%**. (b) I_c = 8 A, I_t = 8.93 A → $m = \sqrt{(2[(I_t/I_c)^2 - 1])} = \sqrt{(2 \times 0.246)} =$ **0.70**. (c) P_c = 100 W, $m = 1 \rightarrow P_t =$ **150 W**, sidebands share 50 W.

3 The AM Family: DSB-SC, SSB, VSB MCQ

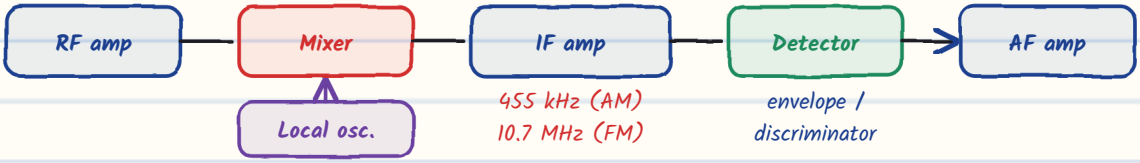


red line = carrier, blue = sidebands, green = bandwidth

Type	Carrier	Sidebands	BW	Saved (m = 1)	Detector	Use
DSB-FC	✓	both	$2f_m$	0	envelope	MW/SW radio
DSB-SC	X	both	$2f_m$	**66.7%**	coherent	stereo L–R, QAM
SSB-SC	X	one	f_m	**83.3%**	coherent	HF radio, FDM
VSB	✓ (TV)	1 + vestige	$f_m + f_v$	–	envelope	analog TV

Exam trap An envelope detector **cannot** demodulate DSB-SC or SSB (no carrier): it needs a local carrier locked in frequency and phase (coherent / synchronous detection). For DSB-SC a phase error θ scales the output by cos θ: zero at 90° (quadrature null).

4 Superheterodyne Receiver, Modulators & Detectors MCQ



superhet: every station is shifted to one fixed IF, where most gain and selectivity live

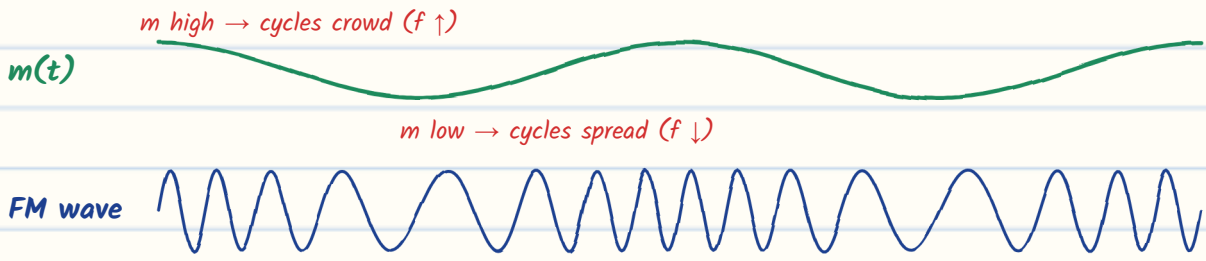
Worked example f_s = 1000 kHz, IF = 455 kHz, high-side LO: $f_{LO} = f_s + IF$ = **1455 kHz**; image = $f_s + 2IF$ = **1910 kHz**, which the RF stage must reject. Envelope detector = diode + RC with $1/f_c \ll RC \ll 1/f_m$ (RC too large → diagonal clipping).

Scheme	Modulator (generation)	Detector (demodulation)
AM (DSB-FC)	collector (high-level), square-law	envelope (diode + RC), square-law
DSB-SC · SSB	**balanced** / ring · filter, phase-shift (Hartley)	coherent (product) detector, Costas loop
FM	varactor + VCO (direct) · Armstrong (indirect)	slope, Foster–Seeley, ratio, **PLL**
PM	varactor phase shifter · NBPM stage	phase detector / PLL

Exam trap A **balanced modulator** gives DSB-SC (carrier suppressed), not ordinary AM. The **ratio detector** rejects amplitude noise itself; Foster–Seeley needs a limiter.

5 FM & PM (Angle Modulation) high-yield

Definition FM: $f_i = f_c + k_f m(t)$. PM: $\varphi(t) = k_p m(t)$. Constant amplitude, so the power $A_c^2/2R$ does not change with modulation.



Formulas peak deviation $\Delta f = k_f A_m$ · $\beta = \Delta f / f_m$ · $s(t) = A_c \cos(2\pi f_c t + \beta \sin 2\pi f_m t)$ · PM: $\beta = k_p A_m$ (independent of f_m)

Carson's rule: $BW \approx 2(\Delta f + f_m) = 2f_m(\beta + 1)$ · NBFM $\beta \ll 1 \rightarrow BW \approx 2f_m$ (like AM) · WBFM $\beta > 1$

Worked examples ✍️

(a) Broadcast FM: $\Delta f = 75$ kHz, $f_m = 15$ kHz → $\beta = 5$, $BW = 2(75 + 15) =$ **180 kHz**.

(b) $10\cos(2\pi \cdot 10^8 t + 5\sin 2\pi \cdot 10^3 t)$: $\beta = 5$, $\Delta f = 5$ kHz, $BW =$ **12 kHz**, $P = 10^2/2 =$ **50 W** ($1\ \Omega$).

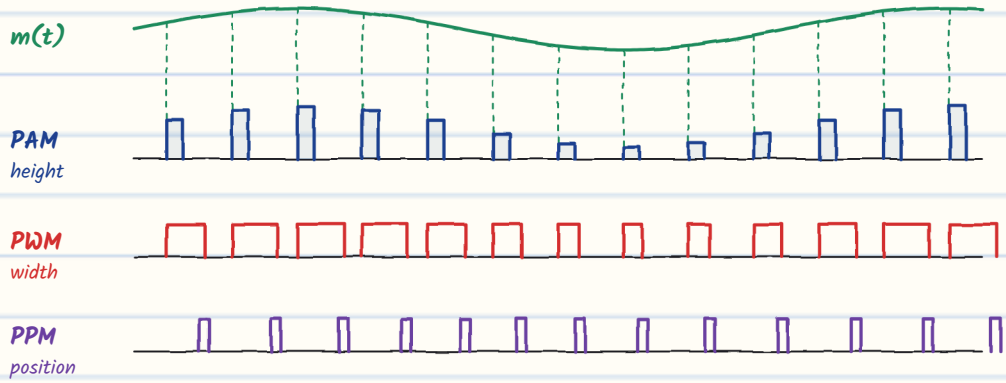
- Spectrum: lines at $f_c \pm nf_m$, amplitudes $A_c J_n(\beta)$ (Bessel); carrier vanishes at $\beta = 2.405$
- FM = PM of the **integrated** message
- Pre-emphasis at Tx (75 µs Americas, 50 µs elsewhere), de-emphasis at Rx

6 AM vs FM

Feature	AM	FM
Bandwidth	$2f_m$; 9 kHz channels (10 kHz in the Americas)	$2(\Delta f + f_m)$; 200 kHz channels
Noise	poor: noise rides on amplitude; simple, long range	**good**: limiter strips it; line-of-sight
Power	≤ 33.3% in sidebands; P_t grows with m	P_t constant, independent of β
Band · IF	MW ≈ 530–1600 kHz · 455 kHz	VHF 88–108 MHz · 10.7 MHz

7 Pulse Modulation: PAM, PWM, PPM MCQ

Definition The carrier is a **pulse train**; one pulse parameter follows each message sample ($f_s \geq 2f_m$). Gaps between pulses carry other channels (**TDM**).



same samples, three ways: pulse height, pulse width, pulse position

Scheme	Varies	Noise	Power / sync	Use
PAM	amplitude	poor	flat-top (sample-and-hold) → aperture effect	first step of PCM
PWM (PDM)	width	better	power varies with width	motor speed, class-D audio
PPM	position	**best of the three**	constant power; needs **sync**	optical, IR remote

8 Pulse Code Modulation (PCM)

written favourite

LPF anti-alias → Sampler $f_s \geq 2f_m$ → Quantiser $L = 2^n$ levels → Encoder n bits/sample → 1011 0010 ...

PCM formulas

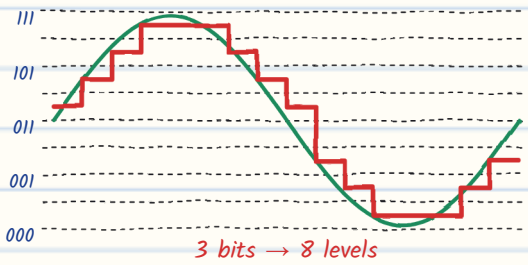
$L = 2^n$ levels · step $\Delta = V_{pp}/L$

max quantisation error = $\pm\Delta/2$ · noise power $\Delta^2/12$

SQNR $\approx 6.02n + 1.76$ dB (full-scale sine)

bit rate $R_b = n \cdot f_s$ (× channels)

min (Nyquist) bandwidth = $R_b/2$



Memory trick "**6 dB per bit**": each extra bit doubles L, halves Δ and adds ≈ 6 dB of SQNR, but also adds f_s bit/s to the bit rate.

→ Worked examples & companding

Worked examples

1. **Telephone voice** (≤ 3.4 kHz), f_s = 8 kHz, n = 8: R_b = 8000 × 8 = **64 kbps** (DS0), L = 256, SQNR = 6.02 × 8 + 1.76 = **49.9 dB**
2. **Audio CD:** 44.1 kHz × 16 bits × 2 channels = **1.4112 Mbps**; SQNR = 6.02 × 16 + 1.76 = **98.1 dB**
3. **Bits for SQNR ≥ 40 dB:** $n \geq (40 - 1.76)/6.02 = 6.35 \rightarrow$ **n = 7**; at f_s = 8 kHz, R_b = 56 kbps, min BW = 28 kHz
4. **V_{pp} = 2 V, n = 3:** L = 8, Δ = 2/8 = 0.25 V, max error = **±0.125 V**

Companding **Compress** at Tx + **expand** at Rx = non-uniform quantisation: fine steps for weak signals, so SQNR stays nearly constant. μ-law (μ = 255): North America, Japan · A-law (A = 87.6): Europe, Bangladesh (E1). Both 8 bits, 64 kbps (ITU-T G.711).

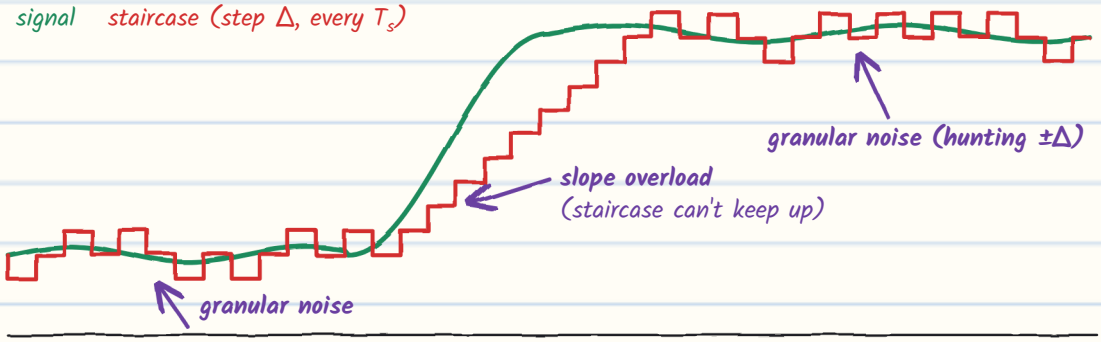
PCM pros: regenerative repeaters (noise does **not** accumulate), encryption, error control, easy TDM. Used in PSTN, CDs, digital TV.
Cons: wide bandwidth (64 kbps vs a 4 kHz analog channel), quantisation noise, needs bit and frame sync.

Exam trap n = bits per sample, $L = 2^n$ levels: 8 bits give **256** levels, not 8. Doubling L adds only 1 bit. PCM was invented by Alec Reeves (1937).

9 Delta Modulation (DM) & Adaptive DM

high-yield

Definition DM sends **1 bit per sample**: 1 if the signal is above the staircase approximation (step up by Δ), 0 if below (step down). Needs heavy oversampling ($f_s \gg 2f_m$); **bit rate = f_s**. Transmitter = comparator + 1-bit quantiser + integrator in the feedback loop; receiver = integrator (accumulator) + LPF.



Rules No slope overload if $\Delta \cdot f_s \geq 2\pi f_m A_m$ (max slope of $A_m \sin 2\pi f_m t$).

Large Δ cures overload but raises **granular noise**; small Δ does the opposite.

Worked example ✍ $A_m = 1$ V, $f_m = 1$ kHz, $f_s = 64$ kHz → $\Delta_{min} = 2\pi \times 1000 \times 1/64000 =$ **0.098 V**; bit rate = **64 kbps**.

ADM: the step size adapts: it grows when successive bits repeat (steep slope) and shrinks when they alternate (flat) → less overload **and** less granular noise, useful for signals with a wide dynamic range. **CVSD** (continuously variable slope DM) is used in Bluetooth voice and military radio. **DPCM** quantises the difference from a predicted sample with several bits; **ADPCM** (ITU-T G.726) gives toll-quality voice at **32 kbps** (DECT cordless phones, VoIP).

Method	Bits/sample	Typical voice rate	Main impairment
PAM	– (analog pulses)	–	channel noise on pulse heights
PCM	n (8)	64 kbps	quantisation noise
DPCM / ADPCM	fewer (4)	32 kbps	smaller quantisation noise; most complex
DM	1	16–64 kbps	slope overload + granular; **simplest**
ADM / CVSD	1	16–32 kbps	both reduced

Exam trap **ADM ≠ ADPCM**: ADM is 1-bit delta modulation with an adaptive step; ADPCM is multi-bit differential PCM with an adaptive quantiser and predictor. The 32 kbps voice codec in phones and VoIP is ADPCM.

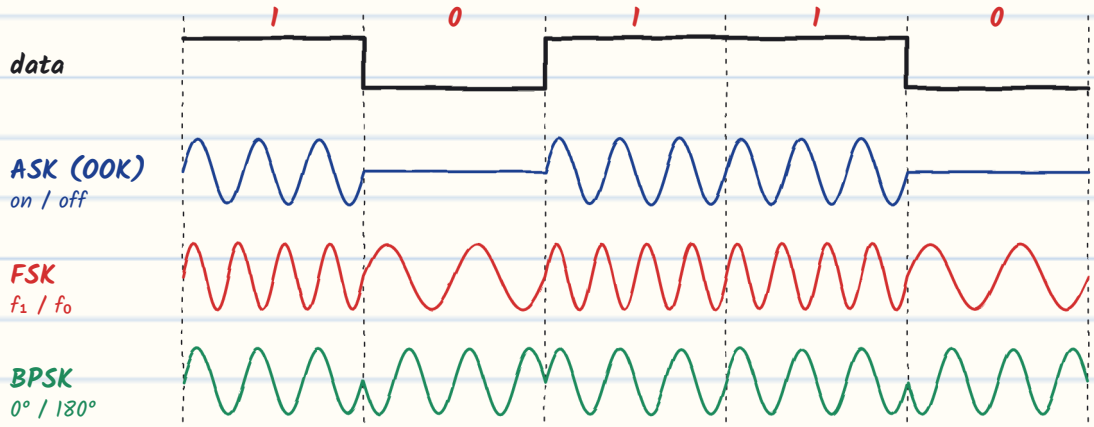
10 Line Coding (Baseband Digital Signals) MCQ

Code	Rule	DC	Clock	Use
NRZ-L / NRZ-I	level per bit; NRZ-I inverts on 1	yes	X	RS-232; USB (NRZI)
Manchester	transition in the middle of every bit	none	✓	10 Mbps Ethernet
Diff. Manchester	mid-bit always; start transition = 0	none	✓	Token Ring
AMI (bipolar)	0 → 0 V; 1s alternate ±	none	X	T1 B8ZS, E1 HDB3

Memory trick **M**anchester = **M**iddle transition = built-in clock at 2× the NRZ bandwidth: 10 Mbps Ethernet = **20 Mbaud**; IEEE: 1 = low → high.

11 Digital Carrier Modulation: ASK, FSK, PSK high-yield

Definition Bits **key** the amplitude (ASK), frequency (FSK) or phase (PSK) of a sine carrier: bandpass transmission (radio, satellite, modems).



BPSK: watch the 180° phase jump at every 1 ↔ 0 change

Scheme	Changes	Noise (BER)	BW	Use
ASK / OOK	amplitude	worst	$2R_b$	fibre, RFID
BFSK	f_1 / f_0	better	$\|f_1 - f_0\| + 2R_b$	modems
BPSK	phase 0°/180°	**best** (+3 dB vs FSK)	$2R_b$	satellite, GPS
DPSK	phase change	≈ BPSK, simpler Rx	$2R_b$	802.11, 1 Mbps
MSK / GMSK	CPFSK, $h = 0.5$	constant envelope	compact	**GSM (GMSK)**

Memory trick Noise ranking = "**P** beats **F** beats **A**" (BW quoted null-to-null): PSK (antipodal, 180° apart) > FSK (orthogonal) > ASK (on/off). Amplitude is what noise attacks most.

Worked example ✍ R_b = 10 kbps

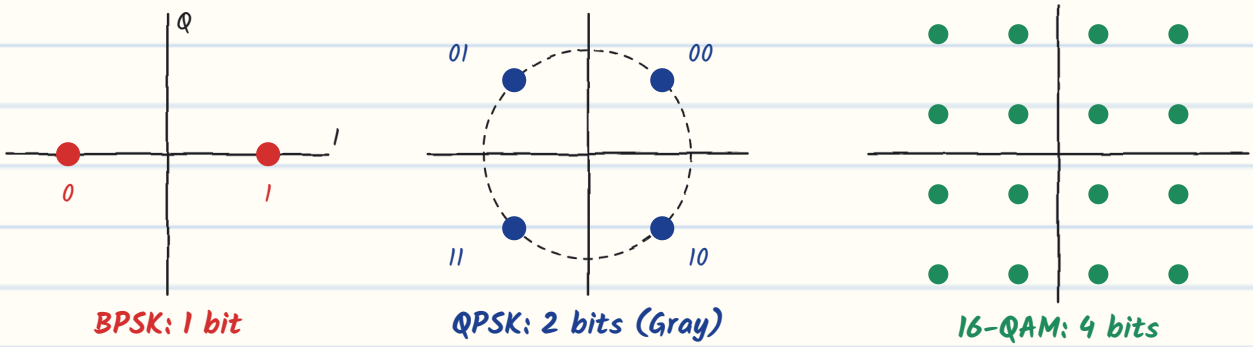
BPSK → null-to-null BW = $2R_b$ = **20 kHz**.

BFSK, $f_1 - f_0$ = 30 kHz → BW = 30 + 20 = **50 kHz**.

12 M-ary Schemes: QPSK, QAM & Constellations

high-yield

Core formulas bits per symbol $k = \log_2 M$ · symbol rate (baud) $R_s = R_b/k$ · min (Nyquist) BW = R_s · ideal spectral efficiency = $\log_2 M$ bit/s/Hz



BPSK: 1 bit QPSK: 2 bits (Gray) 16-QAM: 4 bits

constellations: more points → more bits per symbol, but points crowd together → need higher SNR

Scheme	M	bits/symbol
BPSK	2	1
QPSK (4-QAM)	4	2
8-PSK (EDGE)	8	3
16-QAM	16	4
64-QAM	64	6
256-QAM	256	8
1024-QAM	1024	10

Worked examples ✍

(a) 9600 bps on 16-QAM → R_s = 9600/4 = **2400 baud**

(b) 10 Mbaud on 64-QAM → R_b = 10 × 6 = **60 Mbps**

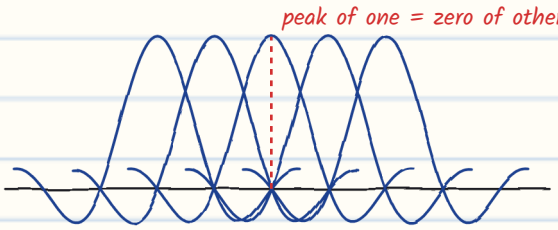
(c) 8 bits per symbol → $M = 2^8$ = **256-QAM**

(d) R_b = 1 Mbps: BPSK needs Nyquist BW 1 MHz, QPSK only **500 kHz** (same BER per bit)

- QAM = ASK + PSK (amplitude and phase); square 16-QAM has 3 amplitude levels and 12 phases. Used in DSL, cable modems, Wi-Fi, 4G/5G
- **Gray coding:** neighbouring points differ in 1 bit, so a symbol error usually costs only 1 bit
- Higher M → more bit/s/Hz but less noise margin; links switch M with SNR (**adaptive modulation**); the ceiling is Shannon's $C = B \log_2(1 + S/N)$

→ OFDM: many slow orthogonal subcarriers

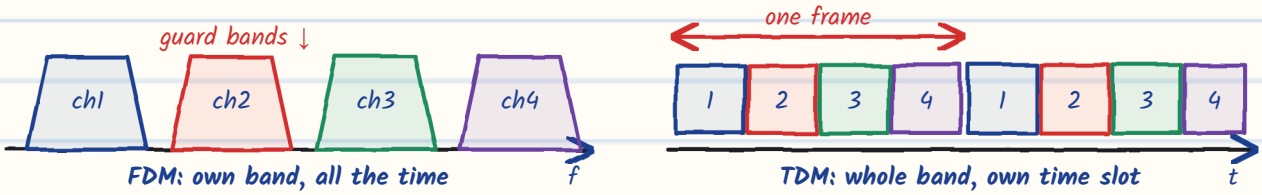
A fast stream is split over N subcarriers spaced $\Delta f = 1/T_{sym}$; each spectrum peaks where all others are zero, so they overlap without interference. Built with **IFFT/FFT**; a **cyclic prefix** absorbs multipath echoes. LTE uses 15 kHz spacing. Used in Wi-Fi (802.11a/g/n/ac/ax), LTE, 5G NR, DVB-T, DAB, ADSL (DMT).



13 Multiplexing: FDM, TDM, WDM

high-yield

Multiplexing vs multiple access **Multiplexing** combines signals at **one place** onto one link (mux → link → demux); **multiple access** lets **separate users** share one medium. Resources: frequency, time, code.



	FDM	TDM	WDM
Shares	frequency	time	wavelength (light)
Separation	guard bands	guard time, framing	DWDM 0.8 nm, CWDM 20 nm
Examples	radio, cable TV	T1/E1, SDH, ISDN	fibre backbones

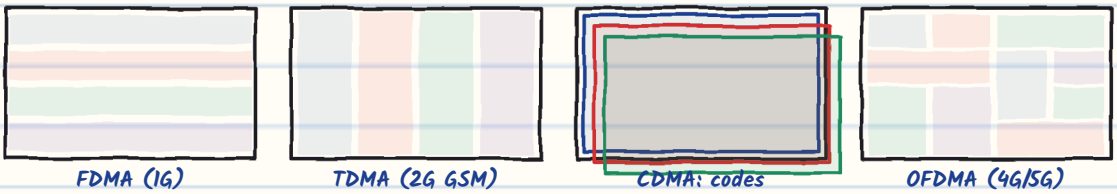
Worked examples ✍ **T1:** 24 × 8 + 1 = 193 bits/frame × 8000 = **1.544 Mbps**. **E1:** 32 slots × 8 × 8000 = **2.048 Mbps** (30 voice + sync + signalling). **FDM:** 5 channels × 100 kHz + 4 guards × 10 kHz = **540 kHz**.

Synchronous TDM: fixed slots, wasted when idle. **Statistical TDM:** slots on demand, with addresses.

T3 = 44.736 Mbps · E3 = 34.368 Mbps · STM-1 = 155.52 Mbps.

14 Multiple Access: FDMA, TDMA, CDMA, OFDMA

high-yield



each box: frequency ↑ vs time →; colours = users. CDMA users overlap fully and are separated by code

Technique	Each user gets · key points	Systems
FDMA	own band, all the time; guard bands	1G AMPS (30 kHz), satellite
TDMA	own time slot on a shared carrier; sync + guard time	GSM = FDMA + TDMA (8 slots / 200 kHz)
CDMA	whole band and time, own spreading code; soft handoff; near-far → **power control**	IS-95 (2G); WCDMA, CDMA2000 (3G); GPS
OFDMA	blocks of subcarriers × time slots; multipath-robust	LTE downlink (SC-FDMA uplink), 5G NR, Wi-Fi 6

Night-Before-the-Exam Sheet

Formula sheet

Antenna ≈ λ/4, λ = c/f

AM: $m = A_m/A_c = (V_{max} - V_{min})/(V_{max} + V_{min})$

AM BW = $2f_m$ · SSB = f_m

$P_t = P_c(1 + m^2/2)$ · $I_t = I_c\sqrt{(1 + m^2/2)}$

$\eta = m^2/(2 + m^2) \leq 33.3\%$

$m_t = \sqrt{(m_1^2 + m_2^2 + \ldots)}$

image = f_s + 2IF · IF 455 kHz / 10.7 MHz

$\beta = \Delta f/f_m$ · $BW = 2(\Delta f + f_m)$

PCM: $L = 2^n$ · $\Delta = V_{pp}/L$ · $R_b = n \cdot f_s$

SQNR ≈ 6.02n + 1.76 dB

DM: $\Delta \cdot f_s \geq 2\pi f_m A_m$ · $R_b = f_s$

$k = \log_2 M$ · $R_s = R_b/k$

T1 = 193 × 8000 = 1.544 Mbps

E1 = 256 × 8000 = 2.048 Mbps

→ Top 12 one-liners for MCQs

1. Standard AM detector: envelope (diode + RC)
2. SSB needs half the bandwidth of DSB
3. VSB: analog TV picture
4. Broadcast FM: Δf = 75 kHz, 200 kHz channels
5. FM carrier vanishes at β = 2.405
6. PPM: best noise immunity of PAM/PWM/PPM
7. Voice PCM = 8 kHz × 8 bits = 64 kbps
8. ADPCM voice = 32 kbps (G.726)
9. Manchester: transition mid-bit, self-clocking
10. QPSK 2, 16-QAM 4, 64-QAM 6 bits/symbol
11. GSM uses GMSK; 8 slots per 200 kHz
12. CDMA: near-far problem → power control

Top traps envelope detector ✗ DSB-SC/SSB · η of AM ≤ 33.3%, not 50% · Carson adds f_m to Δf before doubling · PCM/DM are A/D methods, not carrier modulation · n bits → 2^n levels · ADM ≠ ADPCM · IS-95 is 2G, WCDMA is 3G · GSM is FDMA + TDMA · LTE uplink is SC-FDMA

All tricks together The name says what varies (A/F/P, pulse A/W/P) · "Carson doubles the sum" · "6 dB per bit" · Manchester = Middle transition · "P beats F beats A" · cocktail party: FDMA = separate rooms, TDMA = one room taking turns, CDMA = one room, everyone speaking a different language

Also practise (written): ① AM power for P_c = 400 W, m = 0.8 · ② superhet image frequency · ③ draw Manchester for 10110 · ④ compare FDMA, TDMA and CDMA · ⑤ DM slope-overload condition

✓ Practice MCQs (exam pattern, 10 marks)

try first, then check

1. A 4 kHz message amplitude-modulates a carrier (DSB-AM). The bandwidth is:

(a) 4 kHz (b) 8 kHz
(c) 2 kHz (d) 16 kHz

2. Carrier power 500 W, $m = 0.8$. Total AM power:

(a) 580 W (b) 820 W
(c) 660 W (d) 500 W

3. Maximum transmission efficiency of standard (DSB-FC) AM:

(a) 33.3% (b) 50%
(c) 66.7% (d) 100%

4. FM with $\Delta f = 50$ kHz and $f_m = 10$ kHz; Carson bandwidth:

(a) 100 kHz (b) 60 kHz
(c) 20 kHz (d) 120 kHz

5. PCM with $f_s = 8$ kHz and 7 bits/sample gives a bit rate of:

(a) 64 kbps (b) 8 kbps
(c) 56 kbps (d) 112 kbps

6. Slope overload in delta modulation occurs when:

(a) the step is too small for the signal slope (b) the step is too large for a flat signal
(c) f_s exceeds the Nyquist rate (d) there are too many levels

7. Bits per symbol in 64-QAM:

(a) 8 (b) 4
(c) 64 (d) 6

8. The GSM air interface uses:

(a) pure CDMA (b) FDMA + TDMA
(c) OFDMA (d) pure FDMA

9. An envelope detector can demodulate:

(a) DSB-FC AM (b) DSB-SC
(c) SSB-SC (d) BPSK

10. Line rate of an E1 carrier:

(a) 1.544 Mbps (b) 64 kbps
(c) 2.048 Mbps (d) 44.736 Mbps

Answer key → 1-b · 2-c · 3-a · 4-d · 5-c · 6-a · 7-d · 8-b · 9-a · 10-c

Written question A 4 kHz voice signal is sent by PCM, sampled 25% above the Nyquist rate, with SQNR ≥ 40 dB. Find the bits per sample, levels, bit rate and minimum bandwidth. Why is companding used? 8 marks

1. Nyquist rate = $2 \times 4 = 8$ kHz → $f_s = 1.25 \times 8 =$ **10 kHz** (1)
2. $6.02n + 1.76 \geq 40 \rightarrow n \geq 6.35 \rightarrow$ **$n = 7$ bits**; $L = 2^7 =$ **128 levels**; actual SQNR = 43.9 dB (3)
3. $R_b = n \cdot f_s = 7 \times 10\,000 =$ **70 kbps**; minimum (Nyquist) bandwidth = $R_b/2 =$ **35 kHz** (2)
4. Speech is mostly weak; uniform steps give weak signals poor SQNR. Companding (μ/A-law) uses fine steps for small amplitudes, so SQNR stays nearly constant (2)

→ Before the exam ✓

- ✓ Can derive $P_t = P_c(1 + m^2/2)$ and η
- ✓ Can draw AM, FM, PAM/PWM/PPM, ASK/FSK/PSK
- ✓ Can apply Carson, $n \cdot f_s$, $6.02n + 1.76$
- ✓ Can state the DM slope-overload condition
- ✓ Know TI/EI and 1G–5G access methods